

When Do Covariates Matter? And Which Ones, and How Much?

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Authors often add covariates to a base model sequentially either to test a particular coefficient's "robustness" or to account for the "effects" on this coefficient of adding covariates. This is problematic, due to sequence sensitivity when added covariates are intercorrelated. Using the omitted variables bias formula, I construct a conditional decomposition that accounts for various covariates' role in moving base regressors' coefficients. I also provide a consistent covariance formula. I illustrate this conditional decomposition with NLSY data in an application that exhibits sequence sensitivity. Related extensions include instrumental variables, the fact that my decomposition nests the Oaxaca-Blinder decomposition, and a Hausman test result.

I. Introduction

It is a staple of contemporary empirical research, across many fields of economics, to estimate multiple versions of a linear regression model. One starts from a base specification and then adds progressively more covariates. One basic objective of sequential addition is to evaluate the robust-

I thank Joe Altonji, David Card, Mark Duggan, Bill Evans, Nicole Fortin, Steve Haider, Judy Hellerstein, Kei Hirano, Jon Klick, Thomas Lemieux, Justin McCrary, Ingmar Prucha, Seth Sanders, Jim Smith, Duncan Thomas, Alex Whalley, Justin Wolfers, and seminar participants at Berkeley, the Bureau of Labor Statistics, London School of Economics, University of California Irvine, University of Maryland, and RAND for helpful comments and discussions. I am also very grateful to Jenny Hawkins and Alex Whalley for excellent research assistance. Contact the corresponding author, Jonah Gelbach, at jgelbach@law.upenn.edu. Information concerning access to the data used in this article is available as supplementary material online.

[*Journal of Labor Economics*, 2016, vol. 34, no. 2, pt. 1]

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Submitted June 19, 2012; Accepted August 29, 2014; Electronically published January 22, 2016

ness of the coefficient on some variable of interest, X_1 , as various covariates in X_2 are partialled out. A second common objective is to answer the accounting question: How much of the change in X_1 coefficients can be attributed to different variables in X_2 as one moves from a base specification that has no X_2 covariates to the full specification that has all of them? Many cases of interest involve attempts to determine how much of a cross-group difference in some outcome variables mean is due to group-level heterogeneity in observables, as with the case of the black-white wage gap. In that example, Y represents log wages, X_1 includes a constant and a dummy variable indicating whether a person is black (and possibly also some other baseline controls), and X_2 contains a variety of observed covariates that are potentially related to wages and that may vary systematically across race, such as completed education and Armed Forces Qualification Test (AFQT) score (intended to measure pre-market skills).

The current approach to answering this accounting question is to first measure the change in the estimated race-dummy coefficient as each new set of covariates is added and then attribute this change to the most recently added set of variables. The problem with this approach is that the order in which additional covariates enter the regression can affect the accounting. Of course the problem of sequence dependence is hardly unknown. However, as I discuss in Section II, it is easy to list many otherwise excellent papers in which authors add covariates sequentially in carrying out either robustness or decomposition analyses. In Section III, I illustrate the problem by revisiting Neal and Johnson's (1996) well-known study of the black-white wage gap, showing that using sequential covariate addition gives wildly different results concerning the importance of AFQT and education in explaining the gap. Depending on the sequence in which education and AFQT are added to the wage equation, education either increases the black-white wage gap a little or reduces it substantially. This nonrobustness occurs because of an empirical fact previously noted by Lang and Manove (2006): the black-white education gap depends critically on whether one conditions on AFQT.

Because a robustness measure that is itself not robust to the order of computation is problematic, in Section IV, I develop an alternative, conditional decomposition, based on the simple and well-known least-squares identity that links the estimates of the base- and full-specification coefficients on X_1 via the omitted variables bias formula: $\hat{\beta}_1^{\text{base}} = \hat{\beta}_1^{\text{full}} + (X_1'X_1)^{-1} X_1'X_2\hat{\beta}_2$.¹ An advantage of my decomposition is that it yields consistent estimates of economically and econometrically meaningful population parameters. In the context of the black-white wage gap, this derivation gives a clear meaning to the "effect of adding covariates." Whether variation in a cov-

¹ This identity apparently appeared first in Theil (1957) and has appeared elsewhere over the years, e.g., in Goldberger (1991), as well as, possibly, earlier work by Goldberger.

ariate increases or reduces the gap depends on (a) whether the covariate has a positive effect on wages and (b) whether the covariate has a higher mean among blacks or whites (after partialing out other X_1 variables present in the base specification).² These basic ideas have a long history in econometrics and statistics. But to my knowledge, the decomposition usefulness of the sample omitted variables bias formula has not previously been recognized.³

Since my decomposition is based on parameter estimates computed from the full specification, it is order-invariant. Moreover, it can be implemented by estimating a manageable set of simple auxiliary models and thus requires only minor effort. Implementing my decomposition, I find that—conditional on education, industry dummies, and occupation dummies—heterogeneity in test scores explains roughly a third of the overall black-white wage gap in my National Longitudinal Survey of Youth (NLSY) sample. Education explains less than 10% of the overall gap, variation in occupation of employment explains about 20% of the wage gap, and variation in industry explains virtually none. The results for AFQT and education are quite different from what one would obtain via at least one sequential-addition approach. Thus, my decomposition gives clean conditional answers to questions that previously have hinged on arbitrary choices of sequencing.

² I also show, in Sec. V, that my decomposition nests the Oaxaca-Blinder decomposition. This is useful both because it implies that my Stata code (provided at <https://ideas.repec.org/c/boc/bocode/s457814.html>) can be used to conveniently compute the Oaxaca-Blinder decomposition and because it means that my asymptotic variance formulas, provided in app. B, automatically apply to the elements of the Oaxaca-Blinder decomposition, which should eliminate the practice of providing Oaxaca-Blinder decompositions without estimated standard errors. In addition, I show in Sec. V that under the null hypothesis that X_2 covariates are orthogonal to X_1 , one can use Hausman's (1978) well-known variance-of-differences result to test for statistically significant changes in the coefficient on X_1 . The practical usefulness of this result is that when a single coefficient is of interest, one can assess the null with very little information: only the estimated coefficients and standard errors from the two specifications are needed.

³ One limitation of my decomposition is that its meaningfulness generally requires that the regression function can be correctly written as a linear function of X_1 and X_2 . This implies that none of the variables in X_1 or X_2 is endogenous, mismeasured, or otherwise problematic in a specification that includes both sets of regressors. Such problems can be overcome in the decomposition if they can be overcome in the regression model through the use of valid instruments; I discuss the extension of the decomposition to the instrumental variables context in Subsec. V.A. Regarding nonlinearity, Firpo, Fortin, and Lemieux (2011) discuss a decomposition method for nonlinear situations using the recentered influence function approach of Firpo, Fortin, and Lemieux (2009); see also Fortin, Lemieux, and Firpo (2007). Other decomposition methods for nonlinear situations are suggested by, e.g., Machado and Mata (2005) and Elder, Goddeeris, and Haider (2013). I note also that my interest is in the broad run of cases in which there is no natural ordering in how covariates are causally related to the dependent variable. Altonji et al. (2012) offer a compelling (but likely rare) counterexample, in which the timing of variables suggests a natural ordering in which variables might be added.

II. Examples from the Literature

Consider the case of cross-group gaps in wages. In their outstanding review of work on race, gender, and labor markets, Altonji and Blank (1999) report results from estimating a variety of models relating wages to characteristics of workers and their jobs.⁴ Among their table 4 results are three sets of estimates, models 4–6, of wage regressions using data from the 1995 March Current Population Survey. My table 1 reports their estimates of the coefficient on the black indicator and provides details about included covariates. Altonji and Blank write: “As control variables are added to the model, the negative effect of race . . . on hourly wages becomes less significant. In 1995, black males received 21% lower hourly wages than white males if no control variables were included; they received 12% less once education, experience, and region were controlled for, and they received 9% less when a full set of control variables were included” (3156).

A common inference to draw from this discussion is that education, experience, and region account for $21\% - 12\% = 9\%$ of the black-white wage gap, while additional controls account for $12\% - 9\% = 3\%$. Indeed, the word “account” is sometimes used explicitly to characterize the change in a coefficient of interest when covariates are added (see, e.g., Krueger 1993, 51–52).

Consider also Levitt and Syverson (2008), who use a sample of Chicago-area home sales to test whether agents exploit informational advantages over their principals. They estimate regressions of the log price of home sales on a dummy variable indicating whether the home sold is owned by a real estate agent. When the agent owns the home, the agent is also the principal. A principal-agent problem would cause sale prices of non-agent-owned homes to be lower than those of agents’ own homes. Thus, presence of a principal-agent problem implies a positive coefficient on the agent dummy. My table 2 reports some of Levitt and Syverson’s results from their tables 1 and 2. When only city and year dummies are included, the coefficient on the agent dummy is 12.8 log points; I will consider this the base specification. The agent-dummy coefficient estimate falls to 4.8 when the authors include city-by-year dummies and basic characteristics related to the home’s scale (e.g., number of bedrooms). Levitt and Syverson (2008, 603) write that “the sales price difference between agent-owned and non-agent-owned homes is . . . almost two-thirds less than the within-city difference obtained before conditioning on scale.” The authors then add more detailed house quality indicators, the listing agent’s total number of sales, indicators of various keywords in the home’s listing, and, finally, block dummies. As my table 2 shows, most of the change in the coefficient on the agent dummy between the first (base) specification and the last (full)

⁴ Unless noted otherwise, all references to wages in this paper are to the natural log of wages.

Table 1
Results from Table 4 of Altonji and Blank (1999)

	Altonji and Blank's Model		
	(4)	(5)	(6)
Coefficient on black indicator	-.207	-.117	-.089
Included covariates:			
Hispanic and female indicators	Yes	Yes	Yes
Education, experience, and region	No	Yes	Yes
Occupation, industry, and job characteristics	No	No	Yes

specification occurs when basic house characteristics are added. Levitt and Syverson conclude that “the primary dimension along which agent-owned houses differ from other homes is in terms of scale and readily identifiable amenities . . . including a wide range of controls in addition to these basic ones . . . has a relatively small impact on the measured impact of agent-ownership” (604).

However, it is possible that the scale and basic-amenity variables are correlated with detailed indicators of house quality, description keywords, and block dummies. Thus, it is possible that the coefficient on the agent dummy would move just as much when any of these other sets of covariates is added before adding the basic scale and amenity characteristics. Without a conditional accounting method, it is not possible to assess Levitt and Syverson’s conclusion concerning the cause of the difference in unconditional and conditional agent-dummy effects.

The practice of sequentially adding covariates, whether for purposes of analyzing robustness or accounting for changes in a coefficient of interest, is now common throughout myriad subfields of economics; indeed, it is so pervasive that a full survey of its use is for practical purposes impossible. It has commonly appeared in papers in labor, development, environmental, financial, health, population, and public economics, as well as law

Table 2
Results from Tables 1 and 2 of Levitt and Syverson (2008)

	Table 1, Column 4	Levitt and Syverson's Model			
		Table 2			
		(1)	(2)	(3)	(4)
Coefficient on agent dummy	.128	.048	.042	.038	.037
Included covariates:					
City-by-year interactions	Yes	Yes	Yes	Yes	Yes
Basic house characteristics	No	Yes	Yes	Yes	Yes
Indicators of house quality	No	No	Yes	Yes	Yes
Keywords in description	No	No	No	Yes	Yes
Block fixed effects	No	No	No	No	Yes

and economics and positive political economy.⁵ It is also common in most other disciplines in the empirical social sciences, including political science, sociology, and demography. I emphasize that I do not mean to criticize the work cited here generally; these papers are thoughtful, interesting, and otherwise well executed. My criticism is only with the practice of making inference from sequential covariate addition.

III. Sequential Addition and the Black-White Wage Gap

I now turn to an empirical example involving the black-white wage gap that is based on Lang and Manove's (2006) reconsideration of influential work by Neal and Johnson (1996). Neal and Johnson argue that educational attainment should be excluded from wage equations since schooling decisions may be endogenous to labor market conditions. They use data from the National Longitudinal Survey of Youth 1997 (NLSY79) to evaluate the effects of both excluding schooling and controlling for a person's score on the AFQT, a commonly used measure of ability. Because NLSY subjects generally take the AFQT before they would have finished school, Neal and Johnson argue that this variable is an exogenous measure of ability based on pre-labor market information.

Neal and Johnson find a conventionally large unconditional black-white wage gap, which largely persists when one includes years of schooling as a covariate. In table 3, I report results from wage regressions using my own NLSY sample, which I constructed to roughly match Neal and Johnson's.⁶ In the first row of panel A, I report the estimated coefficient on a race dummy from four different specifications. All specifications include a dummy variable indicating whether a person is Hispanic as well as a linear age term. The specification in the second column also includes years of schooling, the specification in the third column drops schooling but includes a measure of AFQT score, and the specification in the fourth column includes both education and AFQT.

The column 1 coefficient shows that, partialing out age and a Hispanic indicator, the average wage among NLSY young blacks with 1990 earnings is 22.1 log points below the average among young whites. Thus, $\hat{\beta}_{1,\text{race}}^{\text{base}} = -22.1$.

⁵ See, e.g., Bound and Freeman (1992), Card and Krueger (1995, e.g., table 4.10), Kessler and McClellan (1996), Moore, Newman, and Turnbull (1998), Donohue and Levitt (2001), Raphael and Winter-Ebmer (2001), Edlund and Pande (2002), Goldin and Katz (2002), Greenstone (2002), Guryan (2004), Haider et al. (2004), Cullen, Jacob and Levitt (2005), Lee and McCrary (2005), Caceres-Delpiano (2006), Kane, Riegg, and Staiger (2006), Stevenson and Wolfers (2006), Klick and Stratmann (2007), Charles and Guryan (2008), Deaton (2008), Hanushek and Woessmann (2008), Khwaja and Mian (2008), and Lefgren and McIntyre (2009). This list is by no means exhaustive.

⁶ I discuss the sample in appendix A. While my specifications have 156 more observations than do Neal and Johnson's, my results in specifications with given covariate sets are very similar to theirs. I have thus chosen to focus on the sample used here rather than seeking to match the samples more closely.

Table 3
National Longitudinal Survey of Youth Wage Results for Young Men,
Similar to Table 1 of Neal and Johnson (1996)

	A. Black-White Wage Gap Results			
	(1)	(2)	(3)	(4)
Coefficient on black indicator (log points)	-22.1 (3.6)	-19.6 (3.5)	-7.2 (3.8)	-11.7 (3.9)
Included covariates:				
Hispanic indicator,				
and age	Yes	Yes	Yes	Yes
Education	No	Yes	No	Yes
AFQT	No	No	Yes	Yes
B. Implied Decomposition Components				
Sequence in Which Covariates Are Added to Column (1) Specification				
First Add AFQT, Then Add Education First Add Education, Then Add AFQT				
Wage difference attributed to:				
Education completed as of 1990		-2.5	4.5	
AFQT		-7.9	-14.9	
Total difference between (1) and (4)		-10.4	-10.4	

NOTE.—All equations include the same sample used throughout and include 1,749 observations. Estimated standard errors are in parentheses.

Because this estimate is essentially the same as the simple black-white difference in mean log wages when no age control is included and Hispanics are excluded from the sample (21.9 log points), I will refer to $\hat{\beta}_{1,\text{race}}^{\text{base}}$ as simply the black-white wage gap. Adding schooling, in column 2, reduces the black-white wage gap to 19.6 log points; using obvious notation, we thus have $\hat{\beta}_{1,\text{race}}(X_2^{\text{educ}}) = -19.6$.

Neal and Johnson's central result is that when schooling is excluded from the model, adding the AFQT test score as a covariate drives the coefficient on the race dummy much of the way toward zero. My column 3 specification replicates this result. Adding AFQT to the base model in column 1 reduces the black-white wage gap from 22.1 log points to just 7.2 log points: $\hat{\beta}_{1,\text{race}}(X_2^{\text{AFQT}}) = -7.2$. Based on this result, Neal and Johnson argue that most of the observed black-white wage gap is due to pre-market differences across race in ability, whatever the cause of these differences, rather than features of the labor market.

Lang and Manove (2006) construct a signaling model that implies that both educational attainment and test scores must appear as covariates in

wage equations. In column 4 of table 3, I report the estimated black-white wage gap from a specification that includes both AFQT and educational attainment, like the one Lang and Manove estimate.⁷ This specification, which I treat as the full specification for now, yields a black-white wage gap of 11.7 log points. This gap is only about half the size of the column 1 estimate, but it is 4.5 log points greater than Neal and Johnson's preferred specification. For purposes of discussion in the present paper, it will be easiest to accept Lang and Manove's argument that both education and AFQT belong in the wage equation.⁸ Thus, I will regard the column 4 model that includes both AFQT and education as the full specification, so $\hat{\beta}_{1,\text{race}}^{\text{full}} = -11.7$.

The total difference between the base and full specifications is $\hat{\delta}_{\text{race}} = \hat{\beta}_{1,\text{race}}^{\text{base}} - \hat{\beta}_{1,\text{race}}^{\text{full}} = -10.4$ log points. In other words, partialing out both education and AFQT explains away 10.4 log points of the simple column 1 gap of 22.1 log points. In panel B of table 3, I turn to the issue of accounting for this 10.4-point explained part of the wage gap. As discussed above, a common approach to such accounting is to add covariates sequentially and tally up the incremental changes in the black-white wage gap. The entry in the first row and column of panel B shows that adding education to the base model reduces the black-white wage gap by 2.5 log points, which is the difference between the column 1 and column 2 coefficients. The entry in the second row shows that when we add AFQT to the column 2 specification, yielding column 4, the black-white wage gap falls by an additional 7.9 log points. A typical interpretation of these results would be that variation in education explains $2.5/10.4 = 24\%$ of the explained part of the black-white wage gap, while variation in AFQT scores explains $7.9/10.4 = 76\%$.

Of course the education-AFQT sequence is not the only possible one for adding covariates. The second column in panel B of table 3 reports corresponding sequential accounting results when we first add AFQT to the base specification and then add education. Comparing the estimates in col-

⁷ Lang and Manove's sample and use of weights differs somewhat from Neal and Johnson's, but the pattern of estimates they report is similar to the one exhibited in table 3.

⁸ Neal and Johnson's position is that AFQT is a better measure of skill than educational attainment. They argue that any correlation between wages and schooling that holds conditional on AFQT must as a matter of logic be due either to measurement error or endogeneity of post-AFQT administration schooling. Lang and Manove's signaling model provides an alternative, theory-based explanation for the need to include schooling in wage equations. This debate highlights the fact that researchers doing decomposition exercises must believe that the full model is well specified. Empirical researchers typically settle on a full specification that they believe is appropriate (but see Leamer [1985] for a critical view of this fact). The purpose of the present paper is to demonstrate a decomposition method given that such a full model is considered appropriate in the first place. Because it allows me to effect this demonstration, I will accept the Lang and Manove position that both education and AFQT belong in the wage equation.

umns 1 and 3 of panel A now suggests that AFQT variation explains 14.9 log points of the black-white wage gap, for a total of $14.9/10.4 = 143\%$ of the explained part of the black-white wage gap. Meanwhile, when we add education to the column 3 specification, the black-white wage gap increases in magnitude by 4.5 log points. In other words, variation in education now appears to reduce the black-white wage gap, so holding education constant increases the gap.

How can this counterintuitive result occur? As Lang and Manove (2006) comment and demonstrate in their table 2, blacks in the sample have fewer years of schooling than whites, but the sign of this correlation is reversed after we partial out AFQT scores. In table 4, I report results from auxiliary regressions of education on a black indicator. Both specifications include a Hispanic indicator and a linear age control. The specification in column 1 includes no other covariates, and it shows that by 1990, blacks had completed an average of 0.37 fewer years of schooling than whites. In column 2, I add AFQT to the covariate set in this auxiliary model. The results show that, conditional on AFQT, blacks get over a year more education than whites. As I discuss in appendix D, this reversal in the sign of the relationship between educational attainment and race is the key to understanding the sensitivity of the accounting results in table 3 to the sequence in which education and AFQT are added to the base wage specification.

The results in this section are a dramatic demonstration of what has long been known, but which has frequently been ignored in the absence of an alternative approach: that sequential addition of covariates can lead to very different conclusions about how much different covariates contribute depending on their order of addition. Results in table 3 could be read to imply that the black-white differences in education either increase the black-white wage gap by 2.5 log points or reduce it by nearly twice as much. Meanwhile, black-white differences in test scores either reduce the wage gap by three-fourths of the total explained amount, or by more than 140%. Provided one believes that both schooling and test scores belong in the

Table 4
The Black-White Gap in Years of Schooling Completed by 1990s

	(1)	(2)
Coefficient on black indicator (log points)	-.37 (.14)	1.10 (.11)
Included covariates:		
Hispanic indicator and age	Yes	Yes
AFQT	No	Yes

NOTE.—Reported coefficients are estimates of $\Gamma_{\text{race}}^{\text{educ}}$ and $\Gamma_{\text{race}}^{\text{AFQT}}$, computed as the coefficient on a black indicator from least-squares estimation of equations with education/AFQT as the dependent variable and a constant and Hispanic and age controls included. All equations include the same sample used throughout and include 1,749 observations. Estimated standard errors are in parentheses.

wage equation, there is no clear basis on which to choose from these results because the sequence of addition is arbitrary. Yet all of these results have potentially important policy and research consequences.

IV. Accounting for Coefficient Changes

I assume that the two sets of right-hand-side variables, X_1 and X_2 , have k_1 and k_2 variables, respectively. The population linear relationship between Y and $X = [X_1 \ X_2]$ is

$$Y = X_1\beta_1 + X_2\beta_2 + \varepsilon. \quad (1)$$

In the example used in Section III, Y is the log of a person's wage; X_1 contains a constant, a dummy indicating that a person is black, and age; X_2 contains the additional covariates education and AFQT.

With no assumptions on the conditional expectation of ε given the regressors, we can think of (1) as simply specifying a linear projection relationship: assuming all relevant expectations exist, we can define $\beta = (\beta'_1, \beta'_2)'$ as $\beta = E[X'X]^{-1}E[X'Y]$. Under the assumption that $E[\varepsilon_i | X_i] = 0$ for all observations i , β has a causal interpretation as the partial effect of X on Y , since then $E[Y_i | X_i = x] = x'\beta$ and the partial derivative of the conditional expectation is β . In general, I will assume that $E[\varepsilon | X] = 0$ does hold, though none of the results below requires this assumption.⁹

As usual, the ordinary least squares (OLS) estimator for the full vector of partial effects β is given by $\hat{\beta} \equiv (X'X)^{-1}X'Y$. I label $\hat{\beta}_1^{\text{full}}$ and $\hat{\beta}_2$ the components of $\hat{\beta}$ that correspond to the variables in X_1 and X_2 , respectively. Standard results imply that $\hat{\beta}_1^{\text{full}}$ is consistent for β_1 without assuming anything about either β_2 or the correlation between X_1 and X_2 , since all X_2 variation is partialled out in the full specification. Now consider the coefficient on X_1 from the base specification that ignores X_2 . The estimator for this coefficient is $\hat{\beta}_1^{\text{base}} \equiv (X_1'X_1)^{-1}X_1'Y$, whose probability limit is

$$\text{plim } \hat{\beta}_1^{\text{base}} = \beta_1 + \text{plim}(X_1'X_1)^{-1}X_1'X_2\beta_2, \quad (2)$$

$$\text{or } \beta_1^{\text{base}} = \beta_1 + \Gamma\beta_2 = \beta_1 + \delta, \quad (3)$$

where the parameter Γ is the matrix of coefficients from projecting the columns of X_2 on the columns of X_1 ,

$$X_2 = X_1\Gamma + W, \quad (4)$$

and W is a matrix of conformable projection residuals.

Traditionally, (3) is used to demonstrate that δ is the population omitted variables bias from excluding X_2 when estimating β_1 via least squares: β_1^{base} is the parameter identified by this approach, and β_1^{base} differs from β_1 by δ .

⁹ See Wooldridge (2002) for a useful discussion of projections and partial effects.

What has not been realized is that (3) suggests a natural decomposition of the difference in the base- and full-specification estimated coefficients on X_1 . This decomposition is especially useful when β_1^{base} is an interesting parameter in its own right, such as when it represents the unconditional (or age-adjusted) average difference in earnings between races. Empirical decomposition exercises then involve breaking δ into meaningful components and estimating them.

In the example discussed in Section III, X_2 has two columns: one for education and one for AFQT (thus, Γ also must have two columns). In this example, the projection relationship in (4) can also be written, in more detailed form, as

$$X_2^{\text{AFQT}} = \Gamma_0^{\text{AFQT}} + X_{1,\text{race}} \Gamma_{\text{race}}^{\text{AFQT}} + X_{1,\text{Hispanic}} \Gamma_{\text{Hispanic}}^{\text{AFQT}} + X_{1,\text{Age}} \Gamma_{\text{Age}}^{\text{AFQT}} + W^{\text{AFQT}}, \quad (5)$$

$$X_2^{\text{educ}} = \Gamma_0^{\text{educ}} + X_{1,\text{race}} \Gamma_{\text{race}}^{\text{educ}} + X_{1,\text{Hispanic}} \Gamma_{\text{Hispanic}}^{\text{educ}} + X_{1,\text{Age}} \Gamma_{\text{Age}}^{\text{educ}} + W^{\text{educ}}. \quad (6)$$

In this example, X_1 contains a constant, a race indicator, an Hispanic indicator, and age, as in table 3 above. The row of Γ^{AFQT} corresponding to the race dummy tells us the black-white mean difference in residual test scores after partialing out the other elements of X_1 ; the race dummy's row in Γ^{educ} tells us the same thing for years of educational attainment.

To see how the elements of (5) affect the difference between β_1^{base} and β_1^{full} , I rewrite (3) as follows:

$$\beta_1^{\text{base}} - \beta_1 = \Gamma^{\text{AFQT}} \beta_2^{\text{AFQT}} + \Gamma^{\text{educ}} \beta_2^{\text{educ}}, \quad (7)$$

$$\text{so } \delta_{\text{race}} \equiv \beta_{1,\text{race}}^{\text{base}} - \beta_{1,\text{race}} = \Gamma_{\text{race}}^{\text{AFQT}} \beta_2^{\text{AFQT}} + \Gamma_{\text{race}}^{\text{educ}} \beta_2^{\text{educ}}. \quad (8)$$

We can see from (7) how to decompose the explained part of δ_{race} , the simple black-white wage gap:

$$\text{AFQT component: } \delta_{\text{race}}^{\text{AFQT}} = \Gamma_{\text{race}}^{\text{AFQT}} \beta_2^{\text{AFQT}}. \quad (9)$$

$$\text{Education component: } \delta_{\text{race}}^{\text{educ}} = \Gamma_{\text{race}}^{\text{educ}} \beta_2^{\text{educ}}. \quad (10)$$

While I will generalize this claim below, a primary contribution here is to point out that, given the model (1), the parameters in (9) provide a clear answer to the question of how much of the black-white wage gap is explained by variation in education and AFQT. The δ parameters are the mean black-white gap in AFQT or education, scaled by each covariate's wage-equation impact. These covariate mean differences and wage-equation effects are population parameters that do not depend on the order in which covariates are partialled out. Thus, consistent estimates of the δ_{race} param-

eters do not suffer from the sequence sensitivity observed in Section III. Moreover, the δ_{race} parameters are intuitively exactly what researchers seem to mean when they attempt to decompose the causes of the black-white wage gap.¹⁰

In the first row of panel A of table 5, I report estimates of Γ_{race} from my NLSY sample. The estimate in the column headed “Education” shows that after we partial out variation due to Hispanic background and age, blacks have 0.37 fewer years of education (this result also appears in table 4 above). Thus, $\hat{\Gamma}_{\text{race}}^{\text{educ}} = -0.37$. The next row of table 5 reports the estimated coefficients on the education variable from OLS estimation of the full specification given in (1). An additional year of education is associated with an increase in average wages of 4.1 log points, conditional on race, Hispanic background, age, and AFQT. Thus, $\hat{\beta}_2^{\text{educ}} = 0.041$. Equation (9) tells us that the conditional contribution of education is the product of these two estimates: $\hat{\delta}_{\text{race}}^{\text{educ}} = -0.37 \times 0.041$. The next row of the table shows that this product equals -0.0155 : conditional on all the other variables in the model, the black-white wage gap would be roughly 1.6 log points smaller if mean years of schooling were the same across blacks and whites.

I provide analogous results for AFQT in the second column of table 5. Because I have standardized AFQT scores within my sample (see app. A), the coefficient in the first row shows that the average AFQT score among blacks is 0.87 standard deviations lower than among whites, conditional on Hispanic background and age. Thus, $\hat{\Gamma}_{\text{race}}^{\text{AFQT}} = -0.87$. The next row shows that a one standard deviation increase in AFQT is associated with a wage increase of roughly 10 log points: $\hat{\beta}_2^{\text{AFQT}} = 0.1013$. The third row reports the product of these two estimates, $\hat{\delta}_{\text{race}}^{\text{AFQT}} = -0.088$. This estimate tells us that if mean AFQT were the same in this sample and all other relationships were held constant, the black-white wage gap would be 8.8 log points smaller.

Adding together our estimates of these decomposition components shows that variation in education and AFQT together account for a total of 10.4 log points of the black-white wage gap. This figure exactly equals the difference between the column 1 and column 4 specifications of table 3. As I discuss below, this equality is an identity: the sample omitted variables bias formula and the difference between the base- and full-specification coefficient estimates are necessarily the same.¹¹

I now provide general notation for my conditional decomposition. This discussion is simply a formalization of the idea, just used, that the sample analogue of the omitted variables bias formula can serve as the basis

¹⁰ Notice that if there were no mean difference in a covariate across race, then its Γ_{race} coefficient would be zero. In this case, variation in the covariate would explain none of the black-white gap in mean wages. The same would hold if the covariate does not affect the wage, so that $\beta_2 = 0$.

¹¹ I explain what drives the large bias from using the AFQT-education sequential addition approach in detail in appendix D.

Table 5
Understanding the Decomposition of $\hat{\delta}_{\text{race}}$

	Education	AFQT	Total
A. Conditional decomposition components:			
$\hat{\Gamma}_{\text{race}}$	-.3748	-.8698	
$\hat{\beta}_2$	.0413	.1013	
$\hat{\delta}_{\text{race}} (= \hat{\Gamma}_{\text{race}} \times \hat{\beta}_2)$	-.0155	-.0881	-.1036
B. Sequential decomposition components:			
Sequence 1:			
Education-AFQT	-.0249	-.0786	-.1036
Bias	-.0095	.0095	.00
Sequence 2:			
AFQT-Education	.0453	-.1489	-.1036
Bias $(= \hat{\Gamma}_{\text{race}} \times \hat{\beta}_2)$	.0608	-.0608	.0000

NOTE.—Estimates of $\Gamma_{\text{race}}^{\text{educ}}$ and $\Gamma_{\text{race}}^{\text{AFQT}}$ are computed as the coefficient on a black indicator from least-squares estimation of equations with education/AFQT as the dependent variable and a constant and Hispanic and age controls included. Estimates of $\beta_{2,\text{educ}}$ and $\beta_{2,\text{AFQT}}$ are taken from the full specification of the wage equation, which includes all X_1 variables (a constant, black and Hispanic indicators, and age) and both X_2 variables (AFQT and years of schooling). All equations include the same sample used above and include 1,749 observations.

for explaining sensitivity of the estimated coefficient on a vector of base-specification regressors.

Recall that the parameter δ from (3) represents the difference between the population coefficients on in the base and full specifications. Since $\text{plim } \hat{\beta}_1^{\text{base}} = \beta_1 + \delta$, and since $\hat{\beta}_1^{\text{full}}$ is consistent for β_1 , a natural approach to estimating δ is to just use $\hat{\delta} \equiv \hat{\beta}_1^{\text{base}} - \hat{\beta}_1^{\text{full}}$, which is what authors do when they compare base and full specifications. The key algebraic identity driving my decomposition is that the base-full difference in the X_1 coefficient vector identically equals the sample analogue of the omitted variables bias formula:

$$\hat{\delta} = (X_1' X_1)^{-1} X_1' X_2 \hat{\beta}_2. \quad (11)$$

For a very simple proof,¹² observe that the usual least-squares algebra implies

¹² I am grateful to Ingmar Prucha for suggesting this proof. A similar proof using different notation is available in Goldberger (1991, 184). A slightly more detailed alternative proof that makes use of results on orthogonal projections is also available. First, express $\hat{\beta}_2$ as $(X_2' X_2^*)^{-1} X_2' Y$, where $X_2^* \equiv M_1 X_2$ and $M_1 = I - P_1$, with $P_1 = X_1 (X_1' X_1)^{-1} X_1'$ being the projection matrix for X_1 . Thus X_2^* is the part of X_2 that is orthogonal to X_1 . By the Frisch-Waugh-Lovell theorem, $P = P_1 + P_2^*$, where P is the projection matrix using all variables in $X \equiv [X_1' X_2']$, P_1 is the projection matrix for X_1 only, and P_2^* is the projection matrix for X_2^* . Post-multiplying both sides of the projection identity by Y yields $X_1 \hat{\beta}_1^{\text{full}} + X_2 \hat{\beta}_2 = X_1 \hat{\beta}_1^{\text{base}} + X_2^* \hat{\beta}_2$. Rearranging implies $X_1 \hat{\delta} = (X_2 - X_2^*) \hat{\beta}_2 = P_1 X_2 \hat{\beta}_2$. Observing that $X_1' P_1 = X_1'$ and pre-multiplying both sides by $(X_1' X_1)^{-1} X_1'$ then yields $\hat{\delta} = (X_1' X_1)^{-1} X_1' X_2 \hat{\beta}_2 = \hat{\Gamma}_{\text{race}}$, as in (11).

$$Y = X_1 \hat{\beta}_1^{\text{full}} + X_2 \hat{\beta}_2 + \hat{\varepsilon},$$

where $\hat{\varepsilon} = Y - X\hat{\beta}$ is the usual vector of fitted residuals. Now pre-multiply both sides by $(X_1'X_1)^{-1}X_1'$, delivering

$$\hat{\beta}_1^{\text{base}} = \hat{\beta}_1^{\text{full}} + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2,$$

since the fitted residuals are exactly orthogonal to the columns of X_1 by construction of the least-squares estimator. Thus,

$$\begin{aligned}\hat{\delta} &\equiv \hat{\beta}_1^{\text{base}} - \hat{\beta}_1^{\text{full}} \\ &= (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2.\end{aligned}$$

This identity is the key to conditionally decomposing the difference in the base and the full regression coefficients. Since $\hat{\beta}_1^{\text{base}} - \hat{\beta}_1^{\text{full}}$ identically equals the sample omitted variables bias formula, any decomposition of $(X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2$ is also a decomposition of $\hat{\beta}_1^{\text{base}} - \hat{\beta}_1^{\text{full}}$. Let

- X_{2k} be the column of observations on the k^{th} covariate in X_2 ;
- $\hat{\Gamma}_k = (X_1'X_1)^{-1}X_1'X_{2k}$ be the estimated OLS coefficient on X_1 from an auxiliary model with X_{2k} as the dependent variable;
- $\hat{\beta}_{2k}$ be the estimated coefficient on X_{2k} in the full specification of interest.

Then the part of the sample omitted variables bias attributable to covariate k can be written as $(X_1'X_1)^{-1}X_1'X_{2k}\hat{\beta}_{2k} = \hat{\Gamma}_k\hat{\beta}_{2k}$. Because the sample omitted variables bias formula is linear in its k_2 components, it follows that

$$\begin{aligned}\hat{\delta} &= \sum_{k=1}^{k_2} \hat{\Gamma}_k \hat{\beta}_{2k} \\ &= \sum_{k=1}^{k_2} \hat{\delta}_k, \text{ where } \hat{\delta}_k \equiv \hat{\Gamma}_k \hat{\beta}_{2k}.\end{aligned}\tag{12}$$

Equation (12) suggests a straightforward way of carrying out a decomposition:

1. Estimate the full model of interest, which yields the estimated vector $\hat{\beta}_k$.
2. Use OLS to estimate the vector of coefficients on X_1 in a set of auxiliary models with each of the k_2 covariates X_{2k} acting as the dependent variable; this estimate is $\hat{\Gamma}_k$.
3. Multiply $\hat{\Gamma}_k$ by $\hat{\beta}_{2k}$, which yields $\hat{\delta}_k$, the component of the omitted variables bias estimated to be due to each variable k .

Because $\hat{\delta} = \sum_k \hat{\delta}_k$, the estimates of $\hat{\delta}_k$ for groups of covariates can be summed together, that is, into industry, occupation, and so on. Each sum

then yields group-specific omitted variables bias components as desired. For group g , we have $\hat{\delta}^g = \sum_{k \text{ in group } g} \hat{\delta}_k$. The overall difference in coefficients, $\hat{\delta}$, can thus also be written as the sum of group-wise components: $\hat{\delta} = \sum_g \hat{\delta}^g$. A practical drawback to this approach is that there will often be many covariates, so that k_2 is quite large. In such cases, a simple alternative approach is available: rather than estimating k_2 regressions and then creating G group-specific sums, one can sum first and regress later. Define the heterogeneity variable $\hat{H}_{k(i)} = X'_{2k(i)} \hat{\beta}_{2k}$. For any observation i , $\hat{H}_{k(i)}$ tells us how much of i 's wage is estimated to be due to i 's value of the k th covariate, X_{2k} . The practical decomposition I suggest here is based on the obvious equality

$$(X'_1 X_1)^{-1} X'_1 \hat{H}_k = (X'_1 X_1)^{-1} X'_1 X_{2k} \hat{\beta}_{2k} = \hat{\Gamma}_k \hat{\beta}_{2k} = \hat{\delta}_k.$$

Summing over all covariates in group g and using the fact that $(X'_1 X_1)^{-1} X'_1$ does not vary with the covariate index k , we have

$$\begin{aligned} \hat{\delta}^g &= \sum_{k \in \text{group } g} \hat{\delta}_k \\ &= (X'_1 X_1)^{-1} X'_1 \sum_{k \in \text{group } g} \hat{H}_k. \end{aligned}$$

This small manipulation shows that we can calculate the total group- g contribution to $\hat{\delta}$ as follows:

1. Estimate the full model of interest.
2. For each observation i , sum up $\hat{H}_{k(i)} \equiv X'_{2k(i)} \hat{\beta}_{2k}$ over all covariates k in group g ; call the resulting group-level heterogeneity term $\hat{H}_{(i)}^g = \sum_{k \in \text{group } g} \hat{H}_{k(i)}$.
3. Use OLS to estimate the coefficients on X_1 in an auxiliary model in which the dependent variable for observation i is the group-level heterogeneity term $\hat{H}_{(i)}^g$.

The coefficient estimate resulting from the final step above exactly equals $\hat{\delta}^g$, the group g part of the sample omitted variables bias formula. Aside from computational convenience,¹³ this interpretation of the decomposition has a nice intuitive flavor. In the wage gap example, $H_{(i)}$ is the component of i 's wage that is explained by i 's value of the vector of covariates, $X_{2(i)}$. Overall

¹³ Notice that we need estimate only G auxiliary equations, rather than k_2 . In the NLSY example above, this distinction was irrelevant because there were two covariates with each being the only member of its group. Thus, $k_2 = G = 2$ in that example. However, there are many cases in which researchers include large numbers of covariates, especially large sets of dummy variables, in the X_2 covariate set. I will enlarge the covariate set below to illustrate this point.

differences in the simple and conditional wage gaps can then be seen to be due to black-white differences in the average of the heterogeneity term $H_{(i)}$. When X_1 contains no variables other than a race dummy, the part of this difference attributable to variation in group g covariates is the simple cross-group difference in the mean of $H_{(i)}^g$, the wage heterogeneity due to variables in group g . When X_1 also includes other variables, as in the NLSY example I consider in this paper, the relevant parameter is the conditional cross-group difference in $H_{(i)}^g$, after partialing out these other variables.

To conduct inference, we must be able to estimate the asymptotic distribution of $\sqrt{n}(\hat{\delta}^1 - \delta^1, \hat{\delta}^2 - \delta^2, \dots, \hat{\delta}^G - \delta^G)$. In general, all estimators involved in my decomposition are asymptotically normal. Since the decomposition involves continuously differentiable functions of these estimators, joint asymptotic normality of the decomposition components follows from the delta method. Deriving the covariance matrix for the vector of all $\hat{\delta}^g$ estimators is straightforward but lengthy, so I relegate the derivation to appendix B.¹⁴

To illustrate the usefulness of the grouping approach just discussed, I now observe that the NLSY has information on additional potentially important explanatory variables. In particular, I add 12 occupation dummies and 12 industry dummies to the covariate set, so that X_2 now includes 26 variables. I continue to treat the years of education and AFQT variables as their own groups, while adding another group each for occupation and industry. Thus $G = 4$ with this enlarged covariate set.¹⁵

Table 6 reports results from the full specification that includes a constant, race and Hispanic-background dummies, a linear age control, and all 26 X_2 covariates. The black-white wage gap in this full specification is 9.26 log points, so that a total of $\hat{\delta}_{\text{race}} = 12.85$ log points of the base specification's 22.11 log-point gap is explained by the 26 X_2 covariates. The estimated standard error for $\hat{\delta}_{\text{race}}$ is 2.26 log points.

The bottom part of table 6 reports my detailed decomposition of the part of the black-white wage gap that can be explained by each of four covariate sets, conditional on all of them simultaneously. Variation in AFQT

¹⁴ I note, however, that a key requirement for the auxiliary-model approach to work is that at least one element of Γ^g or β_2^g be nonzero. If this condition does not hold, then the variance of $\sqrt{n}(\hat{\delta}^g - \delta^g)$ converges to zero, and estimated test statistics will have nonstandard behavior. For any covariate set such that the researcher allows the possibility that X_1 and X_2^g are uncorrelated, $\Gamma^g = 0$ is possible. To rule out the zero-variance case, it is then important to verify that not all elements of β_2^g are zero. This can be done by estimating the full model and testing the joint null $H_0 : \beta_2^g = 0$ before undertaking the decomposition.

¹⁵ I emphasize that this exercise requires the assumption of correct specification in the full model that includes both X_1 and all 26 variables in X_2 ; see footnote 8 on this point, vis-à-vis the debate between Neal and Johnson (1996) and Lang and Manove (2006) concerning whether even education belongs in the full model.

Table 6
Decomposing the Black-White Wage Gap into AFQT, Education, Occupation, and Industry Components

	Specification		Explained
	Base	Full	
Black-white gap	-22.11 (3.63)	-9.26 (3.81)	-12.85 (2.26)
Covariates:			
AFQT	No	Yes	-7.06 (1.84)
Education	No	Yes	-1.56 (.66)
12 occupation dummies	No	Yes	-4.34 (1.03)
12 industry dummies	No	Yes	.11 (.90)

NOTE.—Both models include Hispanic-background and linear age controls, and both models use the same 1,749 observations. Standard errors are in parentheses (computed under the assumption of conditional sphericity).

explains 7.06 log points, which is more than half of the explained part of the gap and nearly a third of the total gap. On the other hand, this component of the black-white wage gap is less than half of the amount that Neal and Johnson's specification would imply. Variation in educational attainment explains another 1.56 log points. This component does not explain a large share of the gap, though it is precisely estimated. Variation in occupation, as measured by the 12 occupation dummies, explains 4.34 log points of the black-white wage gap. This represents roughly a third of the part of the gap that is explained by the 26 covariates, and about 20% of the overall black-white wage gap.

Variation in industry plays essentially zero role in explaining the black-white wage gap. This result is perhaps surprising, because both wages and race vary systematically across industry. For example, the range of the estimated industry coefficients in the full specification is over 40 log points. Moreover, a test of the null hypothesis that the vector of 12 industry dummies all have coefficients equal to zero results in a p -value of zero up to four decimal places. Similarly, results from a multivariate regression of the 12 industry dummies on race and Hispanic background dummies as well as age yield a p -value of 0.0062 against the null hypothesis that race is unrelated to industry. In sum:

- wages vary systematically across industry, even after partialing out all other variables;
- industry of employment varies systematically across race, conditional on Hispanic background and age;

- yet the black-white wage gap is unaffected by controlling for industry of employment.

This seemingly anomalous result has a simple explanation: the conditional correlation between race and industry is roughly orthogonal to the conditional correlation between wages and industry. That is, even though $\Gamma^{\text{industry}} \neq 0$ and $\beta_2^{\text{industry}} \neq 0$, it is nonetheless approximately true that $\Gamma^{\text{industry}} \beta_2^{\text{industry}} = 0$. As a consequence, wage heterogeneity due to industry variation, H^{industry} , has the same mean among blacks as whites, conditional on Hispanic background and age. This example points to the importance of accounting directly for the effects of X_2 covariates: the simple facts that X_1 and X_2 are correlated and that X_2 enters the model of interest importantly are necessary but not sufficient conditions for the coefficient on X_1 to be affected by partialing out X_2 .

V. Extensions

In this section, I briefly consider three extensions of the analytical results above.

A. Instrumental Variables Estimators

My first extension involves instrumental variables (IV) models. It can be shown that when β_1 from equation (1) is exactly identified, the equivalence between the sample omitted variables bias formula and $\hat{\beta}_1^{\text{base}}$ and $\hat{\beta}_1^{\text{full}}$ holds when all coefficients are estimated using two-stage least squares (2SLS). When β_1 is overidentified, the equivalence does not hold in fixed samples, but it does hold asymptotically, provided all instruments are valid. Thus, differences in IV estimates of β_1 due to addition of exogenous covariates X_2 can be meaningfully decomposed, as with OLS estimates.¹⁶

In Gelbach (2009), I use the same data I used in Gelbach (2002b) to study the effects of public schooling on maternal labor supply. In Gelbach (2002b), I used quarter of birth dummies to instrument for 5-year-old's enrollment in public school. I found that public school enrollment is associated with sizable increases in labor supply among single mothers whose youngest child is age 5. In Gelbach (2009), I use the IV version of the conditional decomposition discussed above to assess the sensitivity of my original results to controlling for covariates. I find that (a) controlling for covariates does not systematically affect the public school enrollment impact, even though (b) different groups of covariates do affect this impact. This is possible because groups of covariates have approximately equal, opposite-signed bias effects, up to sampling variation (as with the industry dummies

¹⁶ See Gelbach (2009) for details. While researchers often adjust the instrument sets across specifications, I take the instrument set as fixed for purposes of this paper.

part of the decomposition considered in Sec. IV above). This is an interesting finding in light of recent work by Buckles and Hungerman (2008), who find that IV estimates of the returns to schooling increase by 20%–50% when they add family background controls to the model, though they do not test for statistical significance. Because my asymptotic variance formulas can be adopted to cover the IV case, the present paper will allow one to answer whether covariates importantly affect IV estimates in applications like estimating the returns to schooling.

B. The Oaxaca-Blinder Decomposition

The second extension concerns the Oaxaca-Blinder decomposition. Suppose we are interested in decomposing the simple mean difference in black and white wages. Let D be a race dummy, and let X be the full set of other explanatory variables, not including a constant. Finally, let $D \times X$ be the interaction of these variables. It is straightforward to show that the explained and unexplained components of the Oaxaca-Blinder decomposition, also known as quantities and prices, can be found using the following procedure:¹⁷

1. Use least squares to estimate the coefficients on a constant, the race dummy, the explanatory variables, and their interaction, yielding $\hat{\beta}_0$, $\hat{\beta}_d$, $\hat{\beta}_x$, and $\hat{\beta}_{dx}$.
2. For each observation i , construct the estimated heterogeneity term $\hat{H}_{(i)}^X \equiv X_{(i)}\hat{\beta}_x$. The explained component of the Oaxaca-Blinder decomposition equals the cross-group difference in mean values of $\hat{H}^X$. This is true because regressing $\hat{H}^X$ on a constant and D yields $\Delta\bar{X}\hat{\beta}_x$, which is the explained component valued using the estimated wage-equation coefficients for the group with $D = 0$.
3. Construct the heterogeneity estimate $\hat{H}_{(i)}^{DX} \equiv (\hat{\beta}_d + \sum_k X_{k(i)}\hat{\beta}_{k,dx})D_i$, where k indexes the variables in X . The unexplained component of the Oaxaca-Blinder decomposition equals the cross-group difference in mean values of $\hat{H}^{DX}$. This is true because regressing $\hat{H}^{DX}$ on a constant and D yields $\hat{\beta}_d + \bar{X}_1 \times \hat{\beta}_{dx}$, where $\bar{X}_1$ is the vector of explanatory variable means for the group with $D = 1$.

This procedure works because (i) the simple difference in mean wages equals the coefficient on a race dummy from the projection of individual wages on a constant and the race dummy, while (ii) the fully interacted model estimated in the procedure's first step yields the estimated coefficients from race-specific wage equations. Together, $\hat{\beta}_0$ and $\hat{\beta}_x$ are the wage-equation coefficient estimates for the group with $D = 0$. Adding those coefficients to $\hat{\beta}_d$ and $\hat{\beta}_{dx}$, respectively, then yields the coefficient estimates

¹⁷ I provide a detailed proof in appendix C.

for the group with $D = 1$. The second two steps in the procedure above are just the auxiliary regression steps used in my groupwise decomposition, with the groups of covariates being X and DX . The only twist is the addition of β_d to the heterogeneity term in the third step.¹⁸

This relationship between the Oaxaca-Blinder decomposition and my decomposition is practically useful. First, my variance formulas can be used to conduct inference on Oaxaca-Blinder decomposition estimates, which is very rarely done.¹⁹ Second, my auxiliary regression method can be used to compute detailed as well as overall Oaxaca-Blinder decompositions. Given the popularity of the Oaxaca-Blinder decomposition, and the interest in questions like “Does industry or occupation explain more of the wage gap in an OBD?” my variance formulas will be very useful for applied practice.²⁰

C. Using the Hausman Test

A third extension involves testing for statistically significant changes in the coefficient on X_1 under the null hypothesis that X_2 covariates are orthogonal to X_1 . It is of course possible to test for orthogonality directly, but this essentially requires estimating a separate least-squares coefficient vector for every covariate in X_2 . In many applications, the dimension of X_2 is large enough that this will be a less than entertaining task. It can be shown that when X_1 is orthogonal to all covariates in X_2 , Hausman’s (1978) variance-of-differences result holds (even though the sufficient conditions

¹⁸ The Oaxaca-Blinder decomposition developed above uses the estimated coefficients for the group with $D = 0$ to measure the explained part of the wage gap. It is well known that the Oaxaca-Blinder decomposition is not unique. To measure the explained part of the wage gap with the other group’s coefficient estimates, one just replaces D with $1 - D$ everywhere above. Other decompositions can also be defined; discussions of this issue appear in Neumark (1988) and Elder, Goddeeris, and Haider (2010), for example.

¹⁹ Horrace and Oaxaca (2001) provide variance formulas that treat sample means as nonstochastic, whereas my more general variance formulas account for this source of covariance.

²⁰ Oaxaca and Ransom (1999) argue that detailed components of $\hat{P}$ suffer from an identification problem due to the fact that adding an arbitrary constant c to the k th variable in X will change its sample mean while leaving its estimated coefficient unchanged. As a consequence, the detailed price component for the group including covariate k will rise by $c\beta_{2,dk}$. Gardeazabal and Ugidos (2004), Yun (2005), and Fortin (2008) suggest various approaches to dealing with the issue raised in Oaxaca and Ransom (1999). My own view, as I argue in Gelbach (2002a), is that the issue itself is not really an identification problem as that term is used in contemporary econometric analysis. Instead, it concerns where in the covariate distribution one chooses to evaluate the effect of differences in coefficients (Jones [1983] makes the Oaxaca and Ransom (1999) point accurately in terms of sensitivity to the choice of baseline, which is another way of putting the issue). I refer interested readers to Gelbach (2002a) for more.

of his theorem need not hold, as Hausman states them). When a single coefficient is of interest, then, one can use already published results to test whether added covariates matter: using reported coefficient estimates and their estimated standard errors, one can construct a test statistic whose asymptotic distribution is standard normal under the null that X_1 and X_2 are orthogonal. Only the estimated coefficients and standard errors from the two specifications are needed to carry out such a test. See Gelbach (2009) for further details.

VI. Conclusion

Comparing coefficient estimates from specifications in which covariates are added sequentially to a base model does not generally identify population parameters of interest. Therefore, this common practice does not generally answer economically or econometrically interesting questions. In this paper, I have shown that the sample version of the omitted variables bias formula can be used to answer the questions that have led researchers to use sequential covariate addition. By contrast to sequential covariate addition, elements of the decomposition I develop here are estimated conditional on all covariates. As such, no sequencing problems arise. Further, my decomposition is easily implemented using intuitively meaningful auxiliary regressions.

I have also presented empirical findings related to the black-white wage gap, using Neal and Johnson (1996) and Lang and Manove (2006) as points of departure. I follow the latter authors and treat education as a necessary covariate in a full wage equation specification. I find that variation in test scores explains 8.8 log points of the overall black-white wage gap; at 40% of the overall gap of 22.1 log points, this is a substantial amount, but it is considerably less than the 14.9 points that would be found by using a sequential-addition approach in which one first adds AFQT to the base specification and then adds education. In an expanded analysis in which the full specification is allowed to include both occupation and industry dummies (i.e., in which they are also treated as exogenous covariates), I find that variation in test scores explain a bit less than a third of the overall wage gap; variation in occupation explains about a fifth; variation in educational attainment explains less than a tenth; and industry variation has no effect on the black-white wage gap, even though industry dummies are significantly correlated with both wages and race. This last result shows that while nonzero values of β_2 and the correlation of X_1 and covariates in X_2 are necessary for a given set of covariates to contribute to the base-full difference in coefficients, they are not sufficient, even together.

I have also discussed a number of extensions. First, my decomposition nests the Oaxaca-Blinder decomposition. This result should be practically useful, since it means my auxiliary-regression approach and variance for-

mulas can be used to compute Oaxaca-Blinder decomposition results. Second, my results carry over exactly to the case of just-identified instrumental variables estimators, and the decomposition method also works in large samples in overidentified cases. Third, cross-specification testing can be done using a Hausman test under the null hypothesis that X_1 and X_2 are uncorrelated, which is sufficient for there to be no difference across specifications in the coefficient on X_1 . As a result, when only one coefficient is of interest, one can carry out specification tests using only the information typically reported in tables provided in published and working papers.

I conclude with one final observation: except in special cases such as the one considered in Altonji, Bharadwaj, and Lange (2012), there really is no reason to sequentially add X_2 covariates to a base model. Sequential addition can obscure, overstate, or understate the true part of δ that can be attributed to variation in any given set of X_2 variables. The only meaningful way to estimate the sensitivity of β_1 to covariates is to add all the covariates at once and then compare $\hat{\beta}_1^{\text{base}}$ and $\hat{\beta}_1^{\text{full}}$. Providing tables with subsets of X_2 added sequentially across columns or rows thus makes little sense, and this practice should simply be abandoned. By contrast, my decomposition method has all the advantages of existing practice. Its only disadvantage has to do with variance computation. Since conventional practice has not included any systematic testing of differences across columns, even this difficulty is not really a disadvantage in context. Nonetheless, I have programmed consistent variance estimators in Stata in publicly available code, so this disadvantage will be irrelevant in practice.

Appendix A

Sample Construction: NLSY Sample

To construct my NLSY sample, I began with all 5,579 males appearing in the nonmilitary NLSY79 sample for 1979. I code as black those who are listed as black and non-Hispanic, and similarly for whites. My educational attainment is the highest grade ever completed as of 1990. The AFQT variable is the sum of the variables ASVAB_SEC2_81, ASVAB_SEC3_81, ASVAB_SEC4_81, and ASVAB_SEC8_81, which I then standardize to have mean 0 and standard deviation 1 in my sample. The wage variable I use is HRP_J1_1990. I use only observations on those aged 18 or younger in 1980.

I form 12 industry dummies from the following 13 groups:

1. Agriculture, Forestry, and Fisheries
2. Mining
3. Construction

4. Durable Manufacturing
5. Nondurable Manufacturing
6. Transportation, Communications, and Other Public Utilities
7. Wholesale and Retail Trade
8. Finance, Insurance, and Real Estate
9. Business and Repair Services
10. Personal Services
11. Entertainment and Recreation Services
12. Professional and Related Services
13. Public Administration

I form 12 occupation dummies from the following 13 groups:

1. Professional, Technical, and Kindred Workers
2. Managers and Administrators, except Farm
3. Sales Workers
4. Clerical and Unskilled Workers
5. Craftsmen and Kindred Workers
6. Operatives, except Transport
7. Transport Equipment Operatives
8. Laborers, except Farm
9. Farmers and Farm Managers
10. Farm Laborers and Farm Foremen
11. Service Workers, except Private Household
12. Private Household Workers
13. Additional NLSY79 Occupation Codes

All models are estimated without sampling weights, following the approach in Neal and Johnson (1996). Estimated standard errors are computed under the assumption of conditional sphericity; heteroskedasticity-robust estimated standard errors were very similar to those reported here.

Appendix B

Covariance Derivation and Estimation

In this appendix, I derive and discuss consistent estimation of the covariance of the vector of decomposition elements. This vector can be written as

$$\hat{\Delta} = \begin{pmatrix} \hat{\delta}^1 \\ \hat{\delta}^2 \\ \vdots \\ \hat{\delta}^G \end{pmatrix} = \begin{pmatrix} \hat{\Gamma}^1 \hat{\beta}_2^1 \\ \hat{\Gamma}^2 \hat{\beta}_2^2 \\ \vdots \\ \hat{\Gamma}^G \hat{\beta}_2^G \end{pmatrix},$$

where we recall that there are k_1 rows in each $\widehat{\delta}^g$, $\widehat{\Gamma}^g$ is $k_1 \times k^g$, and $\widehat{\beta}_2^g$ is a $k^g \times 1$ column vector. Define $\Delta \equiv (\delta^{1'}, \delta^{2'}, \dots, \delta^{G'})'$. Consider the g th component of $\sqrt{n}(\widehat{\Delta} - \Delta)$:

$$\begin{aligned}\sqrt{n}(\widehat{\delta}^g - \delta^g) &= \widehat{\Gamma}^g \sqrt{n}(\widehat{\beta}_2^g - \beta_2^g) + \sqrt{n}(\widehat{\Gamma}^g - \Gamma^g) \beta_2^g \\ &= \widehat{\Gamma}^g \sqrt{n}(\widehat{\beta}_2^g - \beta_2^g) + \left(\frac{X_1' X_1}{n} \right)^{-1} \frac{X_1' v^g}{\sqrt{n}},\end{aligned}\quad (\text{B1})$$

where $v^g = W^g \beta_2^g$ is the population residual from taking the projection relationship between the columns of X_2^g on X_1 and post-multiplying by β_2^g , that is, $X_2^g \beta_2^g = X_1 \Gamma^g \beta_2^g + W^g \beta_2^g$. The asymptotic covariance between $\widehat{\delta}^g$ and $\widehat{\delta}^b$ involves the expectation of the product of (B1) and its transposed analogue when we replace g with b :²¹

$$\begin{aligned}\text{ACov}(\widehat{\delta}^g, \widehat{\delta}^{b'}) &= \text{plim} \left\{ \widehat{\Gamma}^g \sqrt{n}(\widehat{\beta}_2^g - \beta_2^g) \sqrt{n}(\widehat{\beta}_2^b - \beta_2^b)' \widehat{\Gamma}^{b'} \right. \\ &\quad + \left(\frac{X_1' X_1}{n} \right)^{-1} \frac{X_1' v^g v^{b'} X_1}{n} \left(\frac{X_1' X_1}{n} \right)^{-1} \\ &\quad + \widehat{\Gamma}^g \sqrt{n}(\widehat{\beta}_2^g - \beta_2^g) \frac{v^{b'} X_1}{\sqrt{n}} \left(\frac{X_1' X_1}{n} \right)^{-1} \\ &\quad \left. + \left(\frac{X_1' X_1}{n} \right)^{-1} \frac{X_1' v^g}{\sqrt{n}} \sqrt{n}(\widehat{\beta}_2^b - \beta_2^b)' \widehat{\Gamma}^{b'} \right\} \\ &= \Gamma^g \text{ACov}(\widehat{\beta}_2^g, \widehat{\beta}_2^{b'}) \Gamma^{b'} + Q_{11}^{-1} \text{plim} \left(\frac{X_1' v^g v^{b'} X_1}{n} \right) Q_{11}^{-1} \\ &\quad + \Gamma^g \text{ACov} \left[\sqrt{n}(\widehat{\beta}_2^g - \beta_2^g), \frac{v^{b'} X_1}{\sqrt{n}} \right] Q_{11}^{-1} \\ &\quad + Q_{11}^{-1} \text{ACov} \left[\frac{X_1' v^g}{\sqrt{n}}, \sqrt{n}(\widehat{\beta}_2^b - \beta_2^b)' \right] \Gamma^{b'},\end{aligned}\quad (\text{B2})$$

where $Q_{11} = \text{plim}(X_1' X_1 / n)$.

To estimate the four matrices added together on the right-hand side of (B2), it is easiest to think in terms of a stacked system of seemingly unre-

²¹ Note that we can find the asymptotic variance matrix for any $\widehat{\delta}^g$ by setting $g = b$ in (B2).

lated equations. Observe that since $\hat{H}^g \equiv X_2^g \hat{\beta}_2^g$ and $H^g \equiv X_2^g \beta_2^g = X_1 \delta^g + v^g$, we can write $\hat{H}^g = X_1 \delta^g + v^g + X_2^g (\hat{\beta}_2^g - \beta_2^g)$, or $\hat{H}^g = X_1 \delta^g + v^{*g}$, where $v^{*g} \equiv v^g + X_2^g (\hat{\beta}_2^g - \beta_2^g)$. Thus, we have the stacked system

$$\begin{bmatrix} Y \\ \hat{H}^1 \\ \hat{H}^2 \\ \vdots \\ \hat{H}^G \end{bmatrix} = \begin{bmatrix} X & 0 & 0 & \dots & 0 \\ 0 & X_1 & 0 & \dots & 0 \\ 0 & 0 & X_1 & \dots & 0 \\ & & & \ddots & \\ 0 & 0 & 0 & \dots & X_1 \end{bmatrix} \begin{bmatrix} \beta \\ \delta^1 \\ \delta^2 \\ \vdots \\ \delta^G \end{bmatrix} + \begin{bmatrix} \varepsilon \\ v^{*1} \\ v^{*2} \\ \vdots \\ v^{*G} \end{bmatrix}. \quad (\text{B3})$$

Now let $\hat{H}^{\text{stacked}} = (\hat{H}^{1'}, \hat{H}^{2'}, \dots, \hat{H}^{G'})'$ be the $nG \times 1$ column vector of group-level heterogeneity term values (i.e., $\hat{H}^{\text{stacked}}$ is the set of $n \times 1$ fitted $\hat{H}^g$ vectors stacked over the G equations). Let $\theta = (\beta', \Delta')'$ be the $(k_1 + k_2 + Gk_1)$ column vector of population parameters in (B3). The OLS estimator of θ is $\hat{\theta} = (\hat{\beta}', \hat{\delta}^{1'}, \hat{\delta}^{2'}, \dots, \hat{\delta}^{G'})'$, which can be written

$$\hat{\theta} = \begin{bmatrix} (X'X)^{-1} & 0 \\ 0 & \text{I}_G \otimes (X_1'X_1)^{-1} \end{bmatrix} \begin{bmatrix} X'Y \\ (\text{I}_G \otimes X_1')\hat{H}^{\text{stacked}} \end{bmatrix}, \quad (\text{B4})$$

with $(k_1 + k_2 + Gk_1) \times (k_1 + k_2 + Gk_1)$ asymptotic covariance matrix

$$V_{\theta} = \begin{bmatrix} V_{\beta} & C_{\beta\Delta} \\ C_{\Delta\beta} & V_{\text{aux}} \end{bmatrix}. \quad (\text{B5})$$

Here V_{β} is the $(k_1 + k_2) \times (k_1 + k_2)$ covariance matrix for $\hat{\beta}$, the parameters of the full model. For any g and $h \in \{1, 2, \dots, G\}$, the sub-block of this matrix corresponding to the variables in X_2 groups g and h is $V_{\beta}^{gb} = \text{ACov}(\hat{\beta}_2^g, \hat{\beta}_2^h)$ from (B2). This matrix can be estimated consistently by taking the appropriate sub-block of a consistent estimate of the covariance matrix for β , computed as usual. This consistent estimate can then be used in estimating the middle part of the first matrix in (B2), with the outer parts of that matrix estimated using $\hat{\Gamma}^g = (X_1'X_1)^{-1}X_1'X_2^g$ and $\hat{\Gamma}^h = (X_1'X_1)^{-1}X_1'X_2^h$.

With respect to the middle part of the second matrix in (B2), consider the $Gk_1 \times Gk_1$ lower-right block of V_{θ} , given by

$$V_{\text{aux}} = \left[\text{I}_G \otimes \text{plim} \left(\frac{X_1'X_1}{n} \right)^{-1} \right] \text{plim} \left(\frac{(\text{I}_G \otimes X_1')v^*v^{*'}(\text{I}_G \otimes X_1)}{n} \right) \times \left[\text{I}_G \otimes \text{plim} \left(\frac{X_1'X_1}{n} \right)^{-1} \right], \quad (\text{B6})$$

where $v^* \equiv (v^{*1}, v^{*2}, \dots, v^{*G})'$ is the vector of residuals for the bottom G equations of system (B3). The middle part second matrix in (B2),

$\text{plim}(X_1' v^g v^{b'} X_1 / n)$, corresponds to the sub-block of V_{aux} that involves the g th set of rows and b th set of columns (i.e., the rows corresponding to δ^g and columns corresponding to δ^b).

The third matrix in (B2) involves a sub-block of $C_{\beta\Delta}$, while the fourth matrix involves a sub-block of $C_{\Delta\beta} = C_{\beta\Delta}'$. The matrix $C_{\Delta\beta}$ can be formed by stacking $C_{1\beta}$, $C_{2\beta}$, and $C_{G\beta}$, with the g th such $k_1 \times G(k_1 + k_2)$ matrix having the form

$$C_{g\beta} = \text{plim} \left(\frac{X_1' X_1}{n} \right)^{-1} \text{plim} \left(\frac{X_1' v^{*g} \varepsilon' X}{n} \right) \text{plim} \left(\frac{X' X}{n} \right)^{-1}.$$

The part of the fourth matrix in (B2) that involves $\text{ACov}[(X_1' v^g / \sqrt{n}), \sqrt{n}(\hat{\beta}_2^b - \beta_2^b)']$ is given by the columns of $C_{g\beta}$ that correspond to the columns in the b th group of X_2 variables. Analogously, the part of the third matrix in (B2) that involves $\text{ACov}[\sqrt{n}(\hat{\beta}_2^g - \beta_2^g), (v^{b'} X_1 / \sqrt{n})]$ is given by the rows of $C_{\beta g} = C_{g\beta}'$ corresponding to the g th group of X_2 variables.

Estimating the second, third, and fourth matrices in (B2) is straightforward and can be done using the usual sample analogs of relevant parts of $Q \equiv \text{plim}(X' X / n)$ and Γ^g and Γ^b , together with consistent estimates of ε , v^{*g} , and v^{*b} . Consistent estimates of ε can be formed using the usual OLS fitted residual $\hat{\varepsilon} = Y - X\hat{\beta}$ from the full specification of interest. Consistent estimates of v^{*g} can be formed using the fitted residual from the g th auxiliary equation in (B3). As discussed in the main text, we must estimate this equation using $\hat{\beta}_2^g$ in place of the unknown β_2^g . However, $\hat{\beta}_2^g$ is consistent for β_2^g , so nothing is lost as a result—even if we knew β_2^g , we would have to estimate v^g with the consistent estimator $H^g - (X_1' X_1)^{-1} X_1' X_2^g \beta_2^g$. Thus, the feasible fitted residual $\hat{v}^{*g} \equiv \hat{H}^g - (X_1' X_1)^{-1} X_1' X_2^g \hat{\beta}_2^g$ is consistent for v^g since $\hat{\beta}_2^g$ is consistent for β_2^g . The same analysis applies to $\hat{v}^{*b}$. To estimate the middle matrix of the (g, b) block of V_{aux} , we use any consistent estimate for $X_1' \hat{v}^{*g} \hat{v}^{*b'} X_1 / n$, and to estimate $C_{g\beta}$, we use any consistent estimate for $X_1' \hat{v}^{*g} \hat{\varepsilon}' X / n$. Under appropriate iid (independent and identically distributed random variables) assumptions, these matrices can be simplified in familiar ways. When conditional heteroskedasticity or group-level clustering is suspected, the usual robust covariance estimators can be used to estimate them.

Note that in general there is no simple equation-by-equation way to estimate the covariance matrix even for a single group's $\hat{\delta}^g$. For example, consider estimating only the g th equation of system (B3) and then using a standard variance estimator (i.e., the variance estimator one would use if this equation were not an auxiliary equation and did not involve an estimated dependent variable). Because $\hat{v}^{*g}$ is consistent for v^g , this approach is tantamount to ignoring the fact that β_2^g has to be estimated; more formally, it yields only the second matrix in (B2), leaving out the

first, third, and fourth matrices. Fortunately, the Stata code I have written will allow researchers to avoid computing all these matrices by hand.

Finally, when conditional sphericity holds, all the matrices just defined will simplify, since the middle matrix of each then reduces to a scalar multiple of the outer matrices. In this case the limiting variance of $\sqrt{n}(\hat{\beta} - \beta)$ is $V_{\beta} = \sigma_{\varepsilon\varepsilon} Q^{-1}$, where $\sigma_{\varepsilon\varepsilon} = E[\varepsilon_i^2 | X_i]$ and $V_{\text{aux}} = \Sigma_{vv} \otimes (X'X)^{-1}$, where Σ_{vv} is $G \times G$ with scalar gh element $\sigma_{gh} = E[v_i^g v_i^h | X_i]$ for all i . When $\sigma_{g\varepsilon} = E[v_i^g \varepsilon_i | X]$ for all i , $C_{g\beta}$ simplifies to $\sigma_{g\varepsilon} [I_{k_1} \quad \Gamma] Q^{-1}$. Using the partitioned inverse formula, it can be shown that the columns of $C_{g\beta}$ corresponding to $\hat{\beta}_2$ are necessarily zero, so that $\hat{\delta}^g$ and $\hat{\beta}^g$ have zero covariance in this case. Thus under conditional sphericity, only the first and second matrices on the right-hand side of (B2) must be estimated.

Appendix C

The Oaxaca-Blinder Decomposition

Let $\bar{Y}_{\text{black}}$ and $\bar{Y}_{\text{white}}$ be the sample mean of wages among blacks and whites. Let Z be a set of worker characteristics on which we have data (note that Z is not a set of instruments but rather just exogenous covariates). I use Z in this appendix discussion, rather than the notation X used in Subsection V.B, because doing so allows a general connection between the Oaxaca-Blinder decomposition and my decomposition. Let $\hat{\alpha}_{\text{black}} = (\hat{\alpha}_{0,\text{black}}, \hat{\alpha}'_{1,\text{black}})'$ and $\hat{\alpha}_{\text{white}} = (\hat{\alpha}_{0,\text{white}}, \hat{\alpha}'_{1,\text{white}})'$ be the race-specific OLS estimates of coefficients on a constant and Z from separate, race-specific models. Finally, let $\bar{Z}_{\text{black}}$ and $\bar{Z}_{\text{white}}$ be the row vector of sample means of variables in Z for blacks and whites. Because OLS estimates must go through the mean of the data, we have

$$\bar{Y}_{\text{black}} = \hat{\alpha}_{0,\text{black}} + \bar{Z}_{\text{black}} \hat{\alpha}_{1,\text{black}} \quad \bar{Y}_{\text{white}} = \hat{\alpha}_{0,\text{white}} + \bar{Z}_{\text{white}} \hat{\alpha}_{1,\text{white}}.$$

Subtracting $\bar{Y}_{\text{white}}$ from $\bar{Y}_{\text{black}}$ and then adding and subtracting $\bar{Z}_{\text{black}} \hat{\alpha}_{1,\text{white}}$ yields the Oaxaca-Blinder decomposition of mean differences:

$$\bar{Y}_{\text{black}} - \bar{Y}_{\text{white}} = \hat{\alpha}_{0,\text{black}} - \hat{\alpha}_{0,\text{white}} \quad (\text{C1})$$

$$+ \bar{Z}_{\text{black}} (\hat{\alpha}_{1,\text{black}} - \hat{\alpha}_{1,\text{white}}) + (\bar{Z}_{\text{black}} - \bar{Z}_{\text{white}}) \hat{\alpha}_{1,\text{white}}. \quad (\text{C2})$$

Let Δx be the black-white difference in any vector x . Then we can rewrite the Oaxaca-Blinder decomposition as

$$\begin{aligned} \Delta \bar{Y} &= \Delta \hat{\alpha}_0 + \bar{Z}_{\text{black}} \Delta \hat{\alpha}_1 + \Delta \bar{Z} \hat{\alpha}_{1,\text{white}} \\ &= \hat{P} + \hat{Q}, \end{aligned}$$

where $\hat{P} = \Delta\hat{\alpha}_0 + \bar{Z}_{\text{black}}\Delta\hat{\alpha}_1$ and $\hat{Q} = \Delta\bar{Z}\hat{\alpha}_{1,\text{white}}$.

If we think of the $\hat{\alpha}$ coefficients as estimating race-specific hedonic prices of the various Z characteristics, then $\Delta\hat{\alpha} = (\Delta\hat{\alpha}_0, \Delta\hat{\alpha}_1')$ can be thought of as the vector of differences in these prices. As described above, the Oaxaca-Blinder decomposition measures the contribution of the difference in prices via pre-multiplication of $\Delta\hat{\alpha}$ by $[1, \bar{Z}]$. For this reason, the overall term $\hat{P}$ is sometimes called the prices part of the Oaxaca-Blinder decomposition. Similarly, we can think of $\Delta\bar{Z}$ as telling us the cross-race difference in quantities.²²

Now add $\hat{\beta}_1^{\text{full}}$ to both sides of $\hat{\delta} = (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2$, yielding

$$\hat{\beta}_1^{\text{base}} = \hat{\beta}_1^{\text{full}} + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2. \quad (\text{C3})$$

In the wage-gap example, the base specification involves only a constant and a race dummy: $X_1 = [1 \ D]$. Given that a constant is the only other variable in X_1 , $\hat{\beta}_{1D}^{\text{base}} = \bar{Y}_{\text{black}} - \bar{Y}_{\text{white}}$, the cross-race difference in sample means of the dependent variable. The second row of the right-hand side of (C3) is an exact decomposition of this difference in sample means. Therefore, I need only show that the second row of the right-hand side can be written as $\hat{P} + \hat{Q}$.

Let $X_2 = [Z \ DZ]$, where DZ is a matrix of interactions of the race dummy and the vector of characteristics Z (thus $DZ_i = Z_i$ when i is black and 0 when i is white). The matrix of all right-hand side variables is $X = [1 \ D \ Z \ DZ]$, so $Y = X\beta + \varepsilon$ is a fully interacted specification. Let the intercept from this specification be β_{11} , let the coefficient on the race dummy be β_{1D} , and let the vector of coefficients on Z and DZ be β_{2Z} and β_{2DZ} , so that the full specification is

²² The Oaxaca-Blinder decomposition developed above measures the contribution of quantity differences via the term $\hat{Q}$, valuing these differences at the vector of whites' prices, $\hat{\alpha}_{1,\text{white}}$. It is well known that the Oaxaca-Blinder decomposition is not unique. This issue has led to various discussions in the literature about the proper specification of the Oaxaca-Blinder decomposition; see, e.g., Neumark (1988), Fortin (2008), and Elder et al. (2010). Various alternative specifications can be obtained by appropriately constraining the population values corresponding to the $\hat{\alpha}$ coefficients. For example, if we impose the constraint $\alpha_{1,\text{white}} = \alpha_{1,\text{black}} = \alpha_1$, then we could estimate α_1 , $\alpha_{0,\text{white}}$ and $\Delta\alpha_0 \equiv (\alpha_{0,\text{white}} - \alpha_{0,\text{black}})$ by pooling all observations and computing the OLS estimator of the regression of Y on a constant, a dummy indicating that a person is black, and the variables contained in Z (i.e., we would not include interactions of the race dummy and Z variables). Also, we could have used $\bar{Z}_{\text{white}}$ to calculate the price component and for the quantity component (to do so using my approach, one need only use $(1 - D)$ rather than $\bar{D}$ as the race dummy, i.e., use a dummy for being white rather than black, though note the importance of interacting Z and $(1 - D)$ rather than D). For simplicity, here I focus only on the particular Oaxaca-Blinder decomposition described in the text.

$$Y = \beta_{11} + D\beta_{1D} + Z\beta_{2Z} + DZ\beta_{2DZ} + \varepsilon. \quad (C4)$$

Now impose $\beta_{11} = \alpha_{0,\text{white}}$, $\beta_{2Z} = \alpha_{1,\text{white}}$, $\beta_{1D} = \Delta\alpha_0$, and $\beta_{2DZ} = \Delta\alpha_1$, and let $\varepsilon = \varepsilon_{\text{black}} + D(\varepsilon_{\text{black}} - \varepsilon_{\text{white}})$. We can rewrite (C4) as

$$Y = \alpha_{0,\text{white}} + Z\alpha_{1,\text{white}} + D\Delta\alpha_0 + DZ\Delta\alpha_1 + \varepsilon_{\text{black}} + D\Delta\varepsilon,$$

so the OLS coefficients from this specification satisfy the following equalities:

$$\begin{cases} \hat{\beta}_{11} = \hat{\alpha}_{0,\text{white}} \\ \hat{\beta}_{1D}^{\text{full}} = \hat{\alpha}_{0,\text{black}} - \hat{\alpha}_{0,\text{white}} \end{cases} \quad \begin{cases} \hat{\beta}_{2Z} = \hat{\alpha}_{1,\text{white}} \\ \hat{\beta}_{2DZ} = \hat{\alpha}_{1,\text{black}} - \hat{\alpha}_{1,\text{white}} \end{cases} \quad (C5)$$

Next, define covariate groups, indexed with “Z” and “DZ” superscripts to denote the set of covariates included in each group. These groups partition the covariates in X_2 , so we know from the equivalence of $\hat{\beta}_1^{\text{base}} - \hat{\beta}_1^{\text{full}}$ and $\hat{\delta}$ that

$$\hat{\beta}_{1D}^{\text{base}} = \hat{\delta}_D^Z + \hat{\delta}_D^{\text{DZ}} + \hat{\beta}_{1D}^{\text{full}}.$$

Since $\hat{\beta}_{1D}^{\text{base}}$ is the sample mean black-white wage gap, it equals $\hat{P} + \hat{Q}$, and thus

$$\hat{P} + \hat{Q} = \hat{\delta}^Z + \hat{\delta}^{\text{DZ}} + \hat{\beta}_1^{\text{full}}.$$

Thus, there is an immediate equivalence between the sum of the Oaxaca-Blinder decomposition price and quantity components, on the one hand, and the sum of my decomposition’s components and the full model’s race-dummy coefficient, on the other. I will now show that the equivalence is more specific than that: $\hat{\delta}^Z = \hat{P}$, and $\hat{\delta}^{\text{DZ}} = \hat{Q} - \hat{\beta}_1^{\text{full}}$. First, project Z and DZ on $X_1 = [1 \ D]$, yielding

$$(X_1'X_1)^{-1}X_1'Z = \begin{bmatrix} \bar{Z}_{\text{white}} \\ \Delta\bar{Z} \end{bmatrix} \text{ and } (X_1'X_1)^{-1}X_1'DZ = \begin{bmatrix} 0 \\ \bar{Z}_{\text{black}} \end{bmatrix}.$$

The first equality follows because X_1 is just the race dummy, D , and a constant, so the coefficient on the dummy will be the cross-race difference in the mean of Z , with the constant being the average of Z among whites. The second equality follows from the fact that $DZ = 0$ for all whites. Thus the OLS intercept estimate when DZ is the dependent variable is 0, and the coefficient on the race dummy must be the mean of DZ among blacks; since $D = 1$ for all blacks, this mean is $\bar{Z}_{\text{black}}$. Post-multiplying these terms by $\hat{\beta}_{2Z} = \hat{\alpha}_{1,\text{white}}$ and $\hat{\beta}_{2DZ} = \Delta\hat{\alpha}_1$, using the equalities in (C5), and ignoring the row corresponding to the constant yields the groupwise components of my decomposition for $\hat{\beta}_1^{\text{base}}$:

$$\hat{\delta}^Z = \Delta \bar{Z} \hat{\alpha}_{1,\text{white}} \quad \text{and} \quad \hat{\delta}^{DZ} = \bar{Z}_{\text{black}} \Delta \hat{\alpha}_1. \quad (\text{C6})$$

The first equality in (C6) shows that $\hat{\delta}^Z = \hat{Q}$. The second equality shows that $\hat{\delta}^{DZ} = \hat{P} - (\hat{\alpha}_{0,\text{black}} - \hat{\alpha}_{1,\text{white}})$, completing the proof of decomposition equivalence, since $\Delta \hat{\alpha}_0 = \hat{\beta}_{1D}^{\text{full}}$.

Appendix D

Understanding Why the Two Sequential Addition Decompositions Differ

To understand why the two sequential- addition decompositions differ, consider again panel B of table 5, which shows the two sets of accounting estimates reported in table 3. The first set of accounting estimates comes from first adding education to the column 1 base specification, to arrive at column 2 of table 3, and then adding AFQT to that specification to arrive at column 4. Since the columns 1 and 2 estimates of the black-white wage gap are 22.07 and 19.58 log points, this sequence suggests that the black-white wage gap would be 2.49 log points lower if education were held constant, as reported in the first row and column of panel B of table 5. Because the full-specification estimate of the black-white wage gap is 11.71 log points, the difference between columns 2 and 4 is 7.86 log points, as reported in table 3 and repeated in the first row and second column of panel B of table 5.

The $\hat{\delta}_{\text{race}}$ parameters are consistent estimates of the conditional contribution of each covariate. Thus, the bias from using the education-AFQT sequential addition approach equals the difference between the estimates in the first row of panel B and the corresponding $\hat{\delta}_{\text{race}}$ estimates. The second row in panel B shows the resulting bias estimates. These bias estimates sum exactly to zero because my conditional decomposition and the sequential addition decomposition necessarily yield the same total explained gap.

The bias from using the education-AFQT sequence is relatively small in each case—a bit less than one log point. However, the next two rows in panel B show that the reverse sequence causes substantial bias in accounting for the explained part of the gap. While my conditional decomposition shows that variation in educational attainment contributes about 1.5 log points to the black-white wage gap, the AFQT-education sequential addition approach suggests that variation in years of schooling actually reduces the black-white wage gap by 4.5 log points. Moreover, the same bias magnitude applies to AFQT with opposite sign. The AFQT-education sequence of covariate addition would therefore suggest that nearly 15 log points of the black-white wage gap can be attributed to AFQT variation

rather than the consistent estimate of about 9 log points computed when we hold all other variables constant.

What drives the large bias from using the AFQT-education sequential addition approach? Consider the difference between the estimate of -0.0718 in column 3 of table 3, and column 4's full-specification estimate of -0.1171 . The difference between these estimates, $+0.0453$, is the explained-gap component attributed to education using the AFQT-education sequential addition approach. To understand the relationship between this difference and the education component of my conditional decomposition, it will help to consider the following projection relationship:

$$X_2^{\text{educ}} = X_{1,\text{race}} \Gamma_{\text{race}}^{\text{educ(AFQT)}} + X_{1,\text{Hispanic}} \Gamma_{\text{Hispanic}}^{\text{educ(AFQT)}} + X_{1,\text{Age}} \Gamma_{\text{Age}}^{\text{educ(AFQT)}} + X_2^{\text{AFQT}} \Gamma_{\text{AFQT}}^{\text{educ(AFQT)}} + W^{\text{educ(AFQT)}}. \quad (\text{D1})$$

The “educ(AFQT)” superscript notation indicates that the Γ parameters come from an equation relating education to all the X_1 variables, as well as AFQT. Inclusion of AFQT on the right-hand side of (D1) is the only difference between this projection relationship and the one in (5).

Let $\hat{\beta}_{1,\text{race}}(\text{AFQT})$ be the race-dummy estimate from column 3 of table 3, the one that includes AFQT. The equivalence result below can be used to show that

$$\hat{\delta}_{\text{race}}(\text{AFQT}) \equiv \hat{\beta}_{1,\text{race}}(\text{AFQT}) - \hat{\beta}_{1,\text{race}}^{\text{full}} = \hat{\Gamma}_{\text{race}}^{\text{educ(AFQT)}} \hat{\beta}_2^{\text{educ}}.$$

The estimate is the education component of the explained gap computed using the AFQT-education sequential addition approach, which we have seen equals $+0.0453$. We can thus find $\hat{\Gamma}_{\text{race}}^{\text{educ(AFQT)}}$ by dividing $\hat{\delta}_{\text{race}}(\text{AFQT})$ by the table 5 education coefficient in the full-specification wage equation, $\hat{\beta}_2^{\text{educ}} = 0.041$. The result is $\hat{\Gamma}_{\text{race}}^{\text{educ(AFQT)}} = 1.10$, which I earlier reported in the second column of table 4. Relative to my conditional decomposition, the bias in $\hat{\delta}_{\text{race}}(\text{AFQT})$ is thus

$$\begin{aligned} \text{Bias}^{\text{educ(AFQT)}} &= \hat{\delta}_{\text{race}}(\text{AFQT}) - \hat{\delta}_{\text{race}} \\ &= \left[\hat{\Gamma}_{\text{race}}^{\text{educ(AFQT)}} - \hat{\Gamma}_{\text{race}}^{\text{educ}} \right] \times \hat{\beta}_2^{\text{educ}}. \end{aligned} \quad (\text{D2})$$

Equation (D2) shows that the estimated bias in the part of the explained gap attributed to a covariate depends on two factors. One factor is the difference made by conditioning on AFQT when estimating the black-white gap in education, that is, the difference between $\hat{\Gamma}_{\text{race}}^{\text{educ(AFQT)}}$ and $\hat{\Gamma}_{\text{race}}^{\text{educ}}$. In the present example, these projection coefficients are very dissimilar, having different signs and very different magnitudes. Their difference equals $1.10 - (-0.37) = 1.47$. The other factor that affects the bias from using sequential covariate addition to estimate the education share of the explained gap is

$\hat{\beta}_2^{\text{educ}}$, the estimated return to education. While this coefficient is relatively small, its product with 1.47 is substantial, at +6 log points.

Since the biases in the components attributed to AFQT and education must sum to zero, it immediately follows that the sequential addition bias in the component attributed to AFQT when it is added first to the base specification satisfies $\text{Bias}^{\text{AFQT}(\text{AFQT})} = -\text{Bias}^{\text{educ}(\text{AFQT})}$. By symmetry of all definitions above, it can then be shown that the bias in the component attributed to education when it is added first to the base specification satisfies

$$\begin{aligned}\text{Bias}^{\text{AFQT}(\text{educ})} &= \hat{\delta}_{\text{race}}(X_2^{\text{educ}}) - \hat{\delta}_{\text{race}} \\ &= \left[\hat{\Gamma}_{\text{race}}^{\text{AFQT}(\text{educ})} - \hat{\Gamma}_{\text{race}}^{\text{AFQT}} \right] \times \hat{\beta}_2^{\text{AFQT}},\end{aligned}\quad (\text{D3})$$

where $\hat{\Gamma}_{\text{race}}^{\text{AFQT}(\text{educ})}$ is defined analogously to $\hat{\Gamma}_{\text{race}}^{\text{educ}(\text{AFQT})}$: it is the race-dummy coefficient from least-squares estimation of the projection relationship between the left-hand-side variable AFQT and all X_1 variables plus education. This coefficient's value is -0.78 , quite close to the estimate of -0.87 for $\hat{\Gamma}_{\text{race}}^{\text{AFQT}}$. That is, the black-white AFQT gap changes by less than a tenth of a standard deviation when we partial out education in computing this gap. Meanwhile, the return to AFQT from the full-specification wage equation, $\hat{\beta}_2^{\text{AFQT}}$, is roughly 10 log points per standard deviation. That is why the bias of the education component's part of the sequential addition-explained gap is less than one log point when we add education first.

We can summarize this discussion by supposing we have rescaled all X_2 variables so that all wage-equation coefficients, that is, all elements of β_2 , equal one. Whether sequential covariate addition accounting exercises are sensitive to the sequence of addition then depends entirely on the pattern of differences between $\hat{\Gamma}$ projection coefficients estimated with and without conditioning on other X_2 covariates. When the relationship between X_1 and one of the covariates is very sensitive to conditioning on additional covariates, the sequence in which covariates are added will make a large difference in accounting exercises, as in the black-white wage gap.

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